TOMMASO PIGNATA

Mechatronic engineer portfolio

Mechatronic engineer focused on embedded hardware, sensing technologies, and hands-on integration of robotic systems.

London, UK | +44 7709546020 | tommaso.2025@gmail.com
linkedin.com/in/tommaso-pignata-72a304213

Magnetic Tactile Sensor for Robotic Grippers

Problem:

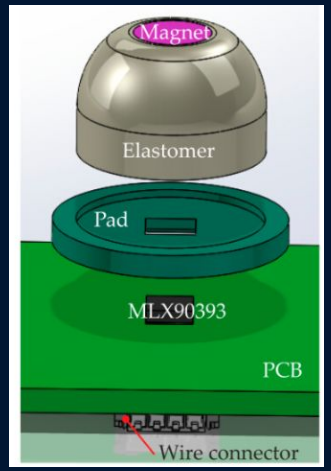
Robotic grippers lack tactile feedback, limiting their ability to detect contact forces and perform reliable object manipulation.

Concept:

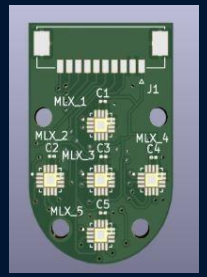
Developed a magnetic-field-based tactile sensing system with high sensitivity using **Hall-effect sensors and permanent magnets embedded within deformable silicone pads.**

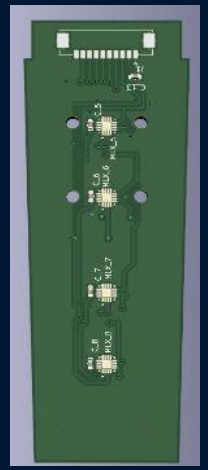
Engineering Work:

- Designed custom PCBs for sensor acquisition
- Integrated Hall-effect sensors with magnet-embedded silicone pad
- Developed firmware for sensor communication (SPI)
- Designed mechanical base for robotic gripper integration
- Designed custom molds for silicone pads
- Created 2 versions for the Allegro hand and Parallel Jaw Gripper finger



Concept





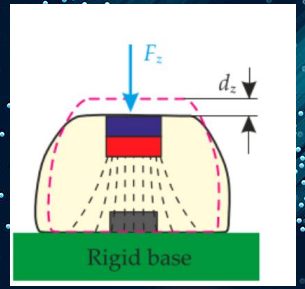
PCB Drawings

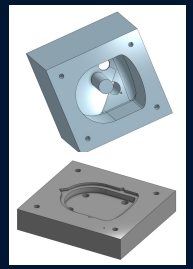
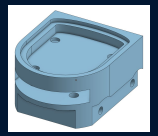
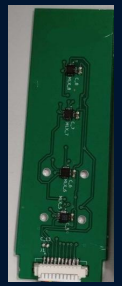
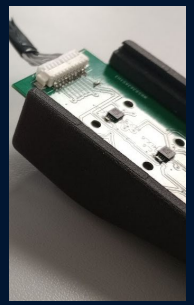
Sensor Architecture and Embedded Hardware

Sensing Principle

Deformation of the silicone pad changes the distance between permanent magnets and Hall sensors, allowing detection of contact forces.

PRINCIPLE



	PCB	MOLD	BASE	ASSEMBLY
ALLEGRO				
J-GRIPPER				

Robotic Integration and Prototype Validation

Prototype Development

- Iterative prototyping of silicone sensing pads
- Integration of sensing elements with robotic fingertips
- Calibration of magnetic response during contact

Outcome

Two tactile sensing prototypes were developed for the Allegro robotic hand and a parallel-jaw gripper. The system reliably detected contact and generated distinct responses for gentle touch, localized point forces, and distributed pressure across the sensing surface.

Jaw-Gripper finger final prototype

Allegro hand finger final prototype

ROS2 Simulation and Motion Planning for KUKA LBR IIWA R800

Developed a complete ROS2 simulation environment for a **7-DOF KUKA LBR IIWA R800 manipulator** integrating **Gazebo Fortress, MoveIt2 motion planning, and ros2_control**.

URDF / Xacro Robot Model

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Gazebo Fortress Simulation

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gz_ros2_control Plugin

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ros2_control Controller Manager

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JointTrajectoryController

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MoveIt2 Motion Planner

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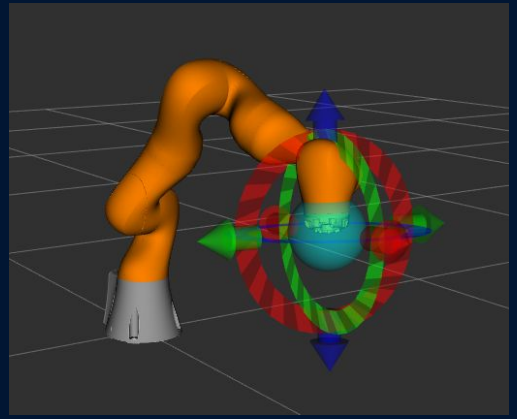
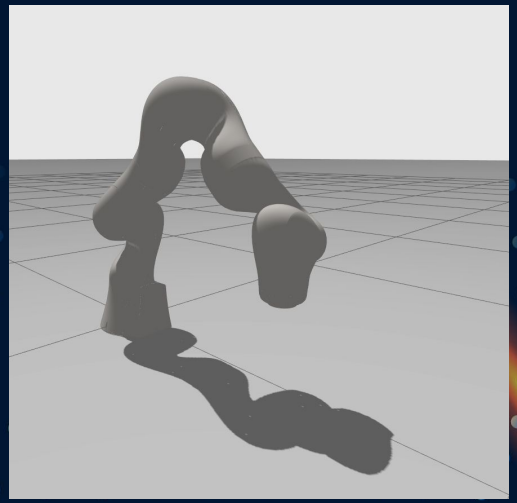
ROS2 Node (Cartesian Waypoints)



RViz model

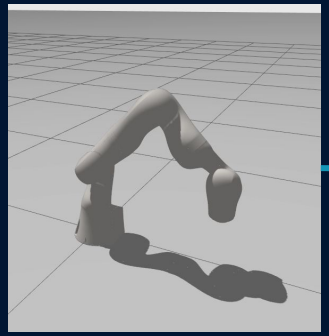


Gazebo model

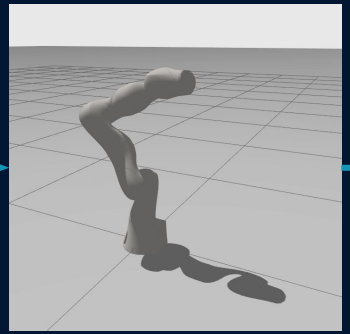
Cartesian Trajectory Planning and Execution

Implementation Highlights

- Fully simulated robotic manipulation pipeline implemented in **Gazebo Fortress** using ROS2
- Custom **ROS2 C++ node** built with MoveIt2 `MoveGroupInterface` to command Cartesian motion
- Cartesian trajectory generation using **`computeCartesianPath()`** with multiple waypoints
- **Forward kinematics implementation** using Denavit–Hartenberg homogeneous transformation matrices
- Pose generation and conversion using **Eigen rotation matrices and quaternions**
- Execution of planned robot trajectories



Start point



Mid point

End point

Battery Inspection Tool – Power Electronics & Embedded Hardware

Problem

Battery modules require periodic inspections to verify **cell condition.**

Existing solutions are often bulky laboratory equipment and unsuitable for **rapid development testing**.

Concept

Develop a **compact electronic inspection tool** capable of performing reliable signal acquisition directly from the Battery Management System.

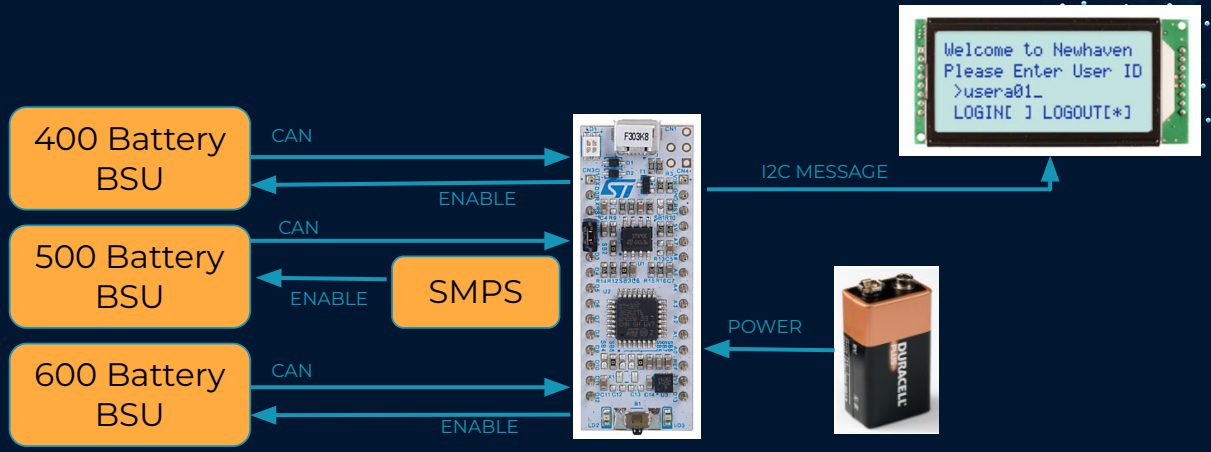
Engineering Work

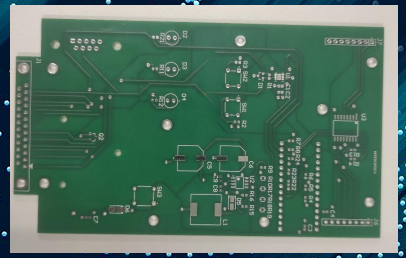
- Installed an **external power architecture** so the tool can work even if the inspected battery is dead.
- **Hardware-based battery identification** without relying on software detection.
- Developed a high-efficiency SMPS boost converter to initialise the 500-series BMS
- Designed the **custom PCB integrating sensing and power circuitry**
- Performed **prototype testing and hardware debugging** to ensure stable operation

Concept

First prototype

Prototype Development & Validation





TOOL PCB

Tool final prototype

Outcome

The final prototype successfully enabled **portable battery inspection and diagnostic measurement**, providing a compact alternative for rapid testing and evaluation.

End-Effector Self-Alignment Sensing System

Problem

Robotic handling requires precise alignment between the end-effector and the target interface. The previous solution relied on an expensive 6-DOF force/torque sensor mounted on the UR10 robot.

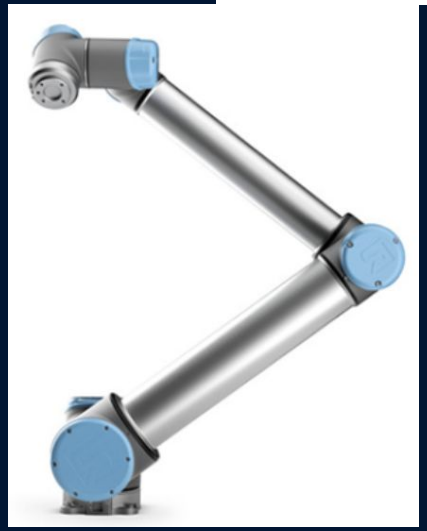
Concept

Develop a cost-effective sensing architecture enabling robotic self-centering using a **Sanwa joystick** and **SICK inductive sensor**.

Engineering Work

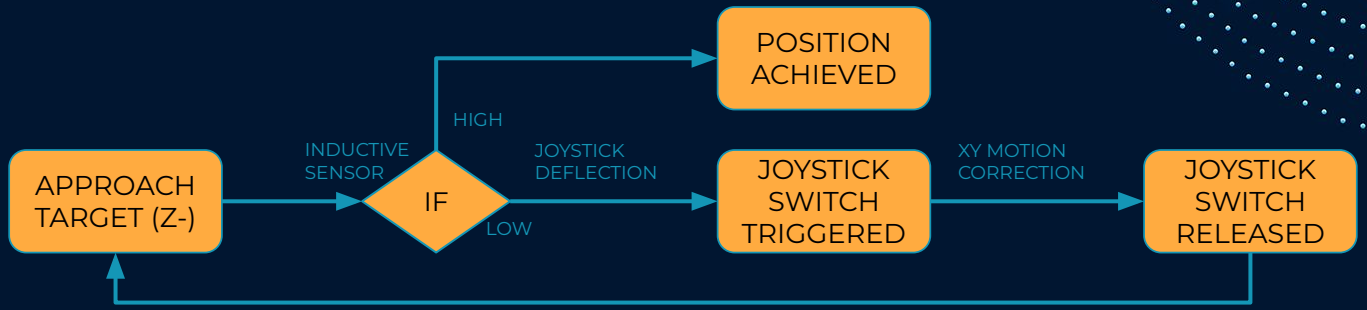
- Engineered a joystick-based XY sensing mechanism with inductive Z-axis detection
- Re-engineered existing components to ensure compatibility with the robotic handling system
- Evaluated multiple design concepts and selected the optimal solution through comparative testing
- Performed validation testing to verify stability and operational viability of the concept

6-DOF SENSOR



UR-10

Prototype Development & Validation



Joystick

Validation

System viability tested across **100 UR10 operational cycles** using controlled **incremental XY misalignment offsets (1–10 mm)** applied in four directions relative to the nominal engagement position.

The self-centering mechanism consistently corrected misalignment with **positioning repeatability of 3σ < 1 mm**.

Outcome

The solution successfully replaced the previous sensor. The system provides reliable alignment while maintaining compatibility with the existing robotic handling infrastructure.

Inductive Sensor